

Arithmetic-Sequence Coordinates for Semiprime Factorization

From Binomial Identities to an Exact Divisor Parameterization, 2-Adic Structure, and Quadratic-Residue Sieving

Experimental Mathematics Notes

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Abstract

We investigate a coordinate system for factoring an odd semiprime $N = pq$, obtained by combining an arithmetic-sequence/binomial transform with a polynomial relation between a near-square coordinate and a sum-gap coordinate. The construction begins with the binomial identities

$$\binom{x}{y} 2^{x-y} = [t^y](2+t)^x, \quad \sum_y \binom{x}{y} 2^{x-y} = 3^x,$$

and their first-moment analogue. These identities convert a quadratic polynomial $P(x, y)$ into a one-variable quadratic $H(x)$, while a second use of the same transform on the diagonal $x + y = z$ linearizes the polynomial exactly. The resulting diagonal parameter $r = s + z$ is not merely correlated with a factor: $r \mid N$ is equivalent to the existence of an integer zero on the corresponding diagonal, and the finite difference of the transformed sequence is exactly r .

The same construction also admits several equivalent descriptions. The y -coordinate reproduces Fermat factorization exactly; the discriminant of P is the squared factor gap $(q - p)^2$. An alternative parametrization by $a = s - p$ and $b = q - s$ converts the problem into the classical relation $ab = s(y + 1) - D$. Reduction modulo powers of two yields an exact parity condition, an involution on the modular roots, and a connection to square roots of N modulo 2^k . These 2-adic structures are exact but, in the experiments considered here, do not by themselves expose the factors. Finally, a quadratic-residue sieve on the Fermat coordinate y can reduce the candidate set by roughly three orders of magnitude, although the observed rank of the true y is quantitatively consistent with ordinary sieve density rather than an additional factor-localizing principle.

The central outcome is therefore a clean algebraic framework with several exact identities, a useful high-strength modular sieve, and a clear separation between identities that merely reparameterize Fermat factorization and mechanisms that would constitute a genuinely new factorization method.

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1 Arithmetic sequences and the binomial identities

The starting point is the weighted binomial sequence

$$W(x, y) = \binom{x}{y} 2^{x-y}, \quad 0 \leq y \leq x, \quad (1)$$

which is naturally generated by the arithmetic shift of the binomial theorem.

1.1 Generating-function interpretation

Since

$$(2 + t)^x = \sum_{y=0}^x \binom{x}{y} 2^{x-y} t^y, \quad (2)$$

we obtain immediately

$$\boxed{\binom{x}{y} 2^{x-y} = [t^y](2 + t)^x.} \quad (3)$$

Consequently,

$$\sum_{y=0}^x \binom{x}{y} 2^{x-y} = 3^x. \quad (4)$$

Differentiating with respect to the auxiliary variable gives the first moment

$$\sum_{y=0}^x y \binom{x}{y} 2^{x-y} = x 3^{x-1}. \quad (5)$$

The second moment, used later as a consistency check, is

$$\sum_{y=0}^x y(y-1) \binom{x}{y} 2^{x-y} = x(x-1) 3^{x-2}, \quad (6)$$

and hence

$$\sum_{y=0}^x y^2 \binom{x}{y} 2^{x-y} = x(x-1) 3^{x-2} + x 3^{x-1}. \quad (7)$$

These formulas imply that any polynomial of degree at most two in the summation variable can be evaluated exactly in closed form under the weight $W(x, y)$. This is the arithmetic-sequence entry point of the construction.

1.2 A general transform identity

For a polynomial

$$Q(x, y) = A(x) + B(x)y + C(x)y^2,$$

its weighted transform

$$\mathcal{T}[Q](x) = \sum_{y=0}^x Q(x, y) \binom{x}{y} 2^{x-y}$$

is

$$\mathcal{T}[Q](x) = 3^{x-2} (9A(x) + 3xB(x) + x(x+2)C(x)). \quad (8)$$

This is simply the substitution of (4)–(7). In particular, no approximation is involved.

2 The semiprime coordinate system

Let

$$N = pq$$

be an odd semiprime with distinct odd primes $p < q$. Define

$$s = \lfloor \sqrt{N} \rfloor, \quad D = N - s^2, \quad (9)$$

and introduce the coordinates

$$\boxed{x = s + 1 - p}, \quad \boxed{y = p + q - 2s - 1}. \quad (10)$$

Equivalently,

$$\boxed{p = s + 1 - x}, \quad \boxed{q = s + x + y}. \quad (11)$$

The quantity $D = N - s^2$ measures the displacement of N above the nearest lower square. The coordinate x measures how far the smaller factor lies below $s + 1$, while y measures the offset of the factor sum from $2s + 1$.

2.1 The polynomial

Substitution of (11) into $N - pq$ gives

$$\boxed{P(x, y) = x^2 + (y - 1)x + (D - s) - (s + 1)y}. \quad (12)$$

The crucial identity is

$$\boxed{P(x, y) = N - (s + 1 - x)(s + x + y)}. \quad (13)$$

At the true coordinates (x_p, y) , we therefore have

$$P(x_p, y) = 0. \quad (14)$$

The same value of y has a second root in x , corresponding to exchanging the factors. Since the coefficient of x is $y - 1$, Vieta gives

$$\boxed{x_p + x_q = 1 - y}. \quad (15)$$

Here

$$x_q = s + 1 - q < 0.$$

Moreover,

$$\boxed{x_p - x_q = q - p}. \quad (16)$$

3 The diagonal: an exact divisor parameterization

The key transformation is to fix the diagonal variable

$$z = x + y. \quad (17)$$

Then $y = z - x$, and substituting into (12) yields

$$P(x, z - x) = x^2 + (z - x - 1)x + (D - s) - (s + 1)(z - x) \quad (18)$$

$$= (z + s)x + D - s - (s + 1)z. \quad (19)$$

Thus

$$\boxed{P(x, z - x) = (z + s)x + D - s - (s + 1)z.} \quad (20)$$

The quadratic term has disappeared completely.

Define

$$r = s + z. \quad (21)$$

Since $D = N - s^2$, the diagonal expression can be rewritten as

$$\boxed{P(x, z - x) = rx + N - r(s + 1).} \quad (22)$$

Equivalently,

$$\boxed{P(x, z - x) = N - r(s + 1 - x).} \quad (23)$$

Theorem 3.1 (Diagonal divisibility equivalence). *For integers x, z , let $r = s + z$. Then*

$$P(x, z - x) = 0 \quad (24)$$

if and only if

$$r \mid N. \quad (25)$$

More precisely, when $r \mid N$, the unique corresponding integer x is

$$\boxed{x = s + 1 - \frac{N}{r}.} \quad (26)$$

Proof. From (23),

$$P(x, z - x) = 0 \iff N = r(s + 1 - x).$$

Thus $r \mid N$, with quotient $N/r = s + 1 - x$. Conversely, any divisor $r \mid N$ inserted into (26) gives an integer zero. $\square$

The diagonal coordinate therefore converts the original two-variable zero problem into a divisor parameterization. At $r = p$ or $r = q$, the corresponding zero is one of the two true roots.

3.1 Exact finite difference

Along a fixed diagonal z , incrementing x by one necessarily decrements y by one. From (20),

$$\boxed{P(x + 1, z - x - 1) - P(x, z - x) = z + s = r.} \quad (27)$$

Hence the diagonal slope is literally the divisor parameter. This fact will reappear in the binomial transform.

4 The binomial transform of the original polynomial

Apply the arithmetic-sequence transform to $P(x, y)$:

$$T(x) = \sum_{y=0}^x P(x, y) \binom{x}{y} 2^{x-y}. \quad (28)$$

Using (4) and (5), one obtains

$$\boxed{T(x) = 3^{x-1} H(x),} \quad (29)$$

where

$$\boxed{H(x) = 4x^2 - (s + 4)x + 3D - 3s.} \quad (30)$$

An equivalent and useful shifted form is

$$\boxed{H(x) = 3N + (x - s - 1)(4x + 3s)}. \quad (31)$$

Therefore, modulo any divisor $r \mid N$,

$$\boxed{H(x) \equiv (x - s - 1)(4x + 3s) \pmod{r}}. \quad (32)$$

At the true factor coordinate $x_p = s + 1 - p$, one obtains

$$\boxed{H(x_p) = p(4p + 3q - 7s - 4)}. \quad (33)$$

4.1 Why ordinary features of H do not locate the factor

The polynomial H is exact, but several natural attempts to use its real geometry were tested experimentally. In a 5000-case study, all cases had real roots. Nevertheless, the true coordinate x_p was far from the roots and the vertex on average, local minima never identified x_p , and simple root/vertex/minimum predictors produced zero correct predictions out of 5000 cases. The exact identity (33) was verified in every test.

The conclusion is important: the transform compresses the two-variable arithmetic into a clean quadratic, but this quadratic's ordinary real geometry does not encode the smaller factor location in a useful way.

5 The binomial transform on a diagonal

The same transform behaves much more rigidly after diagonalization. Fix $r = s + z$, so that

$$P(x, z - x) = rx + N - r(s + 1).$$

Define

$$B_n(r) = \sum_{x=0}^n P(x, z - x) \binom{n}{x} 2^{n-x}. \quad (34)$$

Since the diagonal expression is linear in x , the identities (4) and (5) give

$$B_n(r) = r n 3^{n-1} + (N - r(s + 1)) 3^n \quad (35)$$

$$= 3^{n-1} [3N + r(n - 3(s + 1))]. \quad (36)$$

Hence

$$\boxed{B_n(r) = 3^{n-1} [3N + r(n - 3(s + 1))]} \quad (37)$$

Define the normalized quantity

$$C_n(r) = \frac{B_n(r) - 3^n N}{3^{n-1}}. \quad (38)$$

Then

$$\boxed{C_n(r) = r [n - 3(s + 1)]} \quad (39)$$

and therefore

$$\boxed{C_{n+1}(r) - C_n(r) = r}. \quad (40)$$

Thus the arithmetic-sequence transform does not merely preserve the diagonal slope: after normalization, its first difference is exactly the divisor parameter. Experiments recovered known factors from this transformed difference with 100% success whenever the diagonal radius r was already supplied.

However, scanning candidate radii produced the same search problem as directly testing

$$\gcd(r, N).$$

The transform adds no observed directional information for selecting the correct radius. This distinction is central: *encoding a known divisor more elegantly is not equivalent to discovering the divisor.*

6 The y -coordinate and Fermat factorization

The second coordinate has an equally exact classical interpretation. Treat $P(x, y)$ as a quadratic in x . Its discriminant is

$$\Delta(y) = (y - 1)^2 - 4[(D - s) - (s + 1)y] \quad (41)$$

$$= (2s + 1 + y)^2 - 4N. \quad (42)$$

Therefore

$$\boxed{\Delta(y) = (2s + 1 + y)^2 - 4N.} \quad (43)$$

At the true coordinate $y = p + q - 2s - 1$,

$$2s + 1 + y = p + q, \quad (44)$$

so

$$\boxed{\Delta(y) = (p + q)^2 - 4pq = (q - p)^2.} \quad (45)$$

Thus the exact factor test is simply:

$$(2s + 1 + y)^2 - 4N \quad (46)$$

is a perfect square. This is precisely Fermat factorization in shifted coordinates.

If

$$A = s + \lceil y/2 \rceil,$$

then the classical Fermat quantity is

$$A^2 - N = \frac{(q - p)^2}{4} \quad (47)$$

at the correct y . Experiments showed zero mismatches between the y -search and ordinary Fermat iteration, apart from the expected change of coordinate origin.

The consequence is both positive and negative. Positive: the polynomial framework recovers Fermat factorization exactly and exposes its algebraic structure. Negative: searching y by itself is not a new factorization algorithm.

7 The alternative a, b coordinates

Define

$$a = s - p \geq 0, \quad b = q - s > 0. \quad (48)$$

Then

$$p = s - a, \quad q = s + b. \quad (49)$$

The gap relation becomes

$$q - p = a + b. \quad (50)$$

For the y -coordinate,

$$\boxed{y = b - a - 1,} \quad (51)$$

so

$$\boxed{b - a = y + 1.} \quad (52)$$

Expanding $N = (s - a)(s + b)$ gives

$$D = N - s^2 \quad (53)$$

$$= s(b - a) - ab \quad (54)$$

$$= s(y + 1) - ab. \quad (55)$$

Hence

$$\boxed{ab = s(y + 1) - D.} \quad (56)$$

Equations (52) and (56) determine a and b once y is known. Indeed, letting $d = y + 1$, one has

$$b - a = d, \quad ab = sd - D,$$

and therefore

$$(a + b)^2 = d^2 + 4(sd - D), \quad (57)$$

which is again the discriminant/gap relation in another notation.

Experiments verified all these identities over thousands of random semiprimes. Scanning y and requiring the resulting quadratic to have integral roots yields only the classical square test. The ab reformulation is therefore an exact restatement rather than an independent shortcut.

8 Modular analysis modulo powers of two

The modular structure is considerably richer. Let N be odd and work modulo 2^k . Write

$$P(x, y) = A(x) + B(x)y, \quad (58)$$

where

$$A(x) = x^2 - x + D - s, \quad (59)$$

$$B(x) = x - s - 1. \quad (60)$$

A congruence

$$P(x, y) \equiv 0 \pmod{2^k}$$

has a solution in y exactly when

$$\gcd(B(x), 2^k) \mid A(x).$$

But

$$A(x) = (x - s)(x - s - 1) + D, \quad (61)$$

and modulo any common divisor of $B(x) = x - s - 1$ and 2^k , the term involving $x - s - 1$ vanishes while $A(x) \equiv D$. Since N is odd, the divisibility condition forces $B(x)$ to be odd.

Thus

$$\boxed{P(x, y) \equiv 0 \pmod{2^k} \text{ is solvable in } y \iff x \equiv s \pmod{2}.} \quad (62)$$

When this parity condition holds, $B(x)$ is invertible modulo 2^k , and the corresponding solution is unique:

$$\boxed{y \equiv -(x^2 - x + D - s)(x - s - 1)^{-1} \pmod{2^k}.} \quad (63)$$

Therefore exactly 2^{k-1} residue classes of $x \pmod{2^k}$ admit a corresponding y .

This immediately shows why simply increasing 2-adic precision does not isolate the factor coordinate: half of the x -classes survive for every k .

8.1 The involution

For fixed y , the two roots of $P(x, y)$ satisfy (15). Thus the modular solution set has the exact involution

$$(x, y) \longleftrightarrow (1 - y - x, y). \quad (64)$$

Experiments verified zero involution failures for powers 2^k over multiple precisions. The true pair x_p, x_q always forms one of these two-cycles.

8.2 Fixed points and 2-adic square roots

A fixed point of (64) satisfies

$$x = 1 - y - x,$$

so

$$y = 1 - 2x.$$

Substitution into P gives the exact identity

$$P(x, 1 - 2x) = N - (x - s - 1)^2. \quad (65)$$

Putting

$$r = x - s - 1,$$

we obtain

$$r^2 \equiv N \pmod{2^k}. \quad (66)$$

Thus the fixed points are precisely shifted square roots of N modulo 2^k .

For odd N , these roots exist exactly when $N \equiv 1 \pmod{8}$, in which case there are four roots modulo 2^k for $k \geq 3$. Experiments confirmed the direct correspondence between the fixed points of P and the modular square roots of N .

9 The square-root branches and the factor gap

Let r be any square root of N modulo 2^k :

$$r^2 \equiv N = pq \pmod{2^k}. \quad (67)$$

Then

$$(r - p)(r + p) = r^2 - p^2 \quad (68)$$

$$\equiv pq - p^2 \quad (69)$$

$$= p(q - p) \pmod{2^k}. \quad (70)$$

Hence

$$(r - p)(r + p) \equiv p(q - p) \pmod{2^k}. \quad (71)$$

Similarly,

$$(r - q)(r + q) \equiv q(p - q) \pmod{2^k}. \quad (72)$$

Since p, q, r are odd, both factors on the left have controlled 2-adic valuations. Up to truncation at the modulus, one obtains the exact valuation conservation law

$$v_2(r - p) + v_2(r + p) = v_2(q - p), \quad (73)$$

for the appropriate branch, with the symmetric relation obtained by interchanging p and q .

Experiments found the branch valuations to be permutations of $(1, t-1)$, where $t = v_2(q-p)$, until the modulus cap is reached. This is a genuine and exact 2-adic structure.

However, targeted collision experiments showed that $v_2(q-p)$ is not determined by low-bit fingerprints such as

$$(N \bmod 2^k, s \bmod 2^k)$$

even for moderately large k . Explicit collisions with different values of $v_2(q-p)$ survived targeted searches at $k = 13$ and $k = 14$ across several prime sizes. Therefore the valuation structure cannot be reduced to a simple low-bit fingerprint of (N, s) .

10 Quadratic-discriminant information modulo 2^k

The discriminant relation (43) provides a second modular viewpoint. For the true y ,

$$\Delta(y) = (q-p)^2.$$

Consequently the square residue $\Delta(y) \bmod 2^k$ determines the valuation of $q-p$ exactly whenever the square is nonzero modulo the chosen precision. When the square vanishes modulo 2^k , only the lower bound $2v_2(q-p) \geq k$ is visible.

This is standard 2-adic square information rather than a factorization shortcut. In finite experiments, increasing k quickly makes the valuation exact for most cases, but the same modular data does not determine the actual factor pair.

11 Quadratic-residue sieving of the Fermat coordinate

Although the y -search is exactly Fermat, modular information can still accelerate the search. From (43), any valid factor coordinate must satisfy

$$\Delta(y) = (2s+1+y)^2 - 4N \tag{74}$$

is a quadratic residue modulo every modulus.

For a modulus m , define

$$\mathcal{S}_m = \{u^2 \bmod m : u \in \mathbb{Z}\}. \tag{75}$$

A candidate y survives the corresponding sieve if

$$\Delta(y) \bmod m \in \mathcal{S}_m. \tag{76}$$

The true y necessarily survives every such sieve.

11.1 Local densities

For odd prime moduli, approximately half of the residue classes are quadratic residues. Powers of two are more structured. In the experiments, modulo 8 approximately one half of the candidates survived, while modulo 16 approximately one quarter survived. Combining several small moduli multiplies these local filtering effects to first approximation.

The following table summarizes the observed aggregate reduction in Experiment 58 for 100 random 18-bit-prime semiprimes. The raw number of tested y -values was 397110 across all cases.

Moduli	Survivors	Reduction	Approx. factor
[8]	198555	0.50000000	2.0
[16]	99305	0.74993075	4.0
[16, 3]	48752	0.87723301	8.1
[16, 3, 5]	25147	0.93667498	15.8
[16, 3, 5, 7]	12439	0.96867619	32.0
[16, 3, 5, 7, 11]	6099	0.98464154	65.1
[16, 3, 5, 7, 11, 13]	3067	0.99227670	129.5
[16, 3, 5, 7, 11, 13, 17]	1590	0.99599607	249.8
[16, 3, 5, 7, 11, 13, 17, 19]	812	0.99795523	489.0
[16, 3, 5, 7, 11, 13, 17, 19, 23]	450	0.99886681	882.5

The final wheel therefore removes approximately 99.8867% of the raw candidates, leaving only about one candidate in 882 on average.

11.2 True-coordinate survival

In the same experiment, the true y survived the full modulus set in all 100 cases. This is of course algebraically necessary, but it serves as an implementation check:

$$\boxed{100/100 \text{ true-}y \text{ survival.}} \quad (77)$$

The number of sieve survivors before the true y was small: the observed rank ranged from 1 to 22, with mean 4.50.

11.3 Does the sieve know the true

y ?

A very important distinction arises here. The final sieve density is approximately

$$\rho \approx \frac{450}{397110} \approx 0.00113319. \quad (78)$$

For a candidate interval containing $y + 1$ points, a uniform sieve of density ρ would predict an expected survivor rank of approximately

$$\rho(y + 1). \quad (79)$$

The observed mean rank of 4.50 is of exactly the same scale.

For example, a case with $y = 14073$ had rank 14, while

$$14074\rho \approx 15.95.$$

Another case with $y = 16973$ had rank 16, compared with

$$16974\rho \approx 19.24.$$

Conversely, very small true values such as $y = 147$ or $y = 29$ often produced rank 1 simply because too few candidates existed before the first survivor.

The present evidence therefore supports the conservative interpretation:

The quadratic-residue sieve is a strong candidate reduction mechanism, but the observed rank of the true y is quantitatively consistent with ordinary sieve density. No independent evidence has yet been established that the sieve preferentially selects the factor-producing y .

12 Experiment log and exact results

This section records the principal computational experiments in a compact mathematical form.

Experiment 23: transformed quadratic H

The arithmetic-sequence transform produced $T(x) = 3^{x-1}H(x)$ with H given by (30). The exact factor-coordinate evaluation (33) was verified. A single zero of H occurred in the tested data ($N = 6649 = 61 \cdot 109$, $s = 81$), showing that algebraic degeneracies can occur but are not sufficient to isolate a factor.

Experiments 37–40: low-bit fingerprints and solvability

Attempts to recover the factor coordinate from $(N \bmod 2^k, s \bmod 2^k)$ showed increasing apparent uniqueness at small sample sizes, but later exhaustive/targeted collision tests destroyed universal low-bit claims. Exact algebra established the true criterion (62): solvability modulo 2^k is equivalent to $x \equiv s \pmod{2}$, leaving exactly half of all x -classes.

Experiments 41–43: 2-adic root sets

Enumeration of modular solutions verified that the true coordinate survives at every tested precision. The solution set carries the exact involution (64); its fixed points are exactly the shifted square roots (66).

Experiments 44–47: square-root valuations

The relation between square-root branches and $q - p$ was established exactly through (71)–(73). However, targeted collision tests showed that the valuation of the factor gap is not determined by low-bit fingerprints of (N, s) .

Experiment 48: discriminant identity

The corrected discriminant formula (43) was verified for 200/200 cases. Recovering the exact factor gap from the discriminant square was likewise successful whenever 2-adic precision was sufficient to avoid the zero-square ambiguity.

Experiment 49: equivalence with Fermat

The y -coordinate search matched Fermat factorization exactly. The mapping $A = s + \lceil y/2 \rceil$ introduces only a coordinate shift.

Experiment 50: geometry of H

Across 5000 cases, all transformed quadratics had real roots, but the real roots, vertex, and local-minimum features did not locate x_p . Simple predictors scored zero correct cases. This rules out the most obvious real-analytic interpretations of H .

Experiment 51: exact diagonal parameterization

The identities (20)–(27) were tested over 200 random cases. The linearization, divisor equivalence, and diagonal slope identities had zero failures. The diagonal is therefore an exact divisor parameterization, not a heuristic approximation.

Experiments 52–53: transformed diagonal search

The transformed diagonal sequence $B_n(r)$ was checked against its closed form and against direct gcd detection. All identities held. The transform recovered known factors when r was already supplied, but searching r was equivalent to scanning divisors directly; no new selection principle emerged.

Experiments 55–56: a, b coordinates

The equations (51) and (56) were verified on thousands of random semiprimes. Searching y and enforcing integral reconstruction of a, b remained equivalent to a Fermat square test.

Experiment 57: small-modulus discriminant sieve

The sieve based on quadratic residues of $\Delta(y)$ preserved the true y in every case. Combining 16, 3, 5, 7, 11, 13 reduced the full candidate workload by about 99.3%, from 334556 raw candidates to approximately 2323 survivors in the tested sample.

Experiment 58: extended wheel

Using [16, 3, 5, 7, 11, 13, 17, 19, 23], 100 random cases produced 397110 raw candidates and only 450 survivors, a reduction of 99.8867%. The true y survived in all 100 cases, with sieve rank between 1 and 22 and mean 4.50. This is consistent with the global survivor density and therefore does not, by itself, imply additional factor localization.

13 What is exact, what is classical, and what remains open

It is useful to separate the findings into three categories.

13.1 Exact structural results

The following identities are exact consequences of the coordinate construction:

1. The weighted binomial identities (4)–(7).
2. The polynomial identity (13).
3. Diagonal linearization (20).
4. Divisor equivalence (26).
5. Exact diagonal slope (27).
6. The transformed quadratic identity (29).
7. The transformed diagonal difference identity (40).
8. The discriminant identity (43) and its factor-gap specialization (45).
9. The a, b equations (52) and (56).
10. The exact 2-adic solvability criterion (62).
11. The involution (64) and its fixed-point square-root representation (66).
12. The valuation relations (71)–(73).

13.2 Exact but classically equivalent factorization procedures

The following do not presently constitute new factorization algorithms:

1. Searching y and testing whether $\Delta(y)$ is a square: this is Fermat factorization.
2. Solving for a, b from y : this reduces to the same quadratic discriminant.
3. Searching the diagonal parameter r and using the transformed difference: this reduces to divisor testing or $\gcd(r, N)$.
4. Studying the ordinary roots or vertex of H : the experiments do not indicate a useful factor-location rule.

13.3 Potentially useful computational improvement

The quadratic-residue sieve is the clearest algorithmic improvement discovered in this line of work. It does not change the mathematics of Fermat factorization, but it can reduce the number of expensive exact-square tests by orders of magnitude. For the tested 18-bit-prime cases and the modulus set $[16, 3, 5, 7, 11, 13, 17, 19, 23]$, the reduction was about $885\times$.

The open question is whether the sieve can be organized into a fast wheel whose construction cost is negligible compared with the resulting savings, and whether larger or specially chosen moduli produce a nontrivial improvement beyond the approximately multiplicative density model.

14 Further directions suggested by the structure

The present work narrows the search space for a genuinely new idea. Several directions are especially natural.

14.1 Normalized rank testing

For a case with true coordinate y , define the observed survivor rank $R(y)$ and compare

$$Q = \frac{R(y)}{(y+1)\rho}, \tag{80}$$

where ρ is the empirically calibrated sieve density. If Q is stable around one, this strongly supports the conclusion that the sieve carries no extra directional information.

14.2 Structured wheel generation

The full least-common-modulus construction can become unwieldy. A practical implementation can instead construct a moderate base wheel, for example using

$$16 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = 240240,$$

then apply additional modular filters for 17, 19, 23, $\dots$ to the base residue classes. This avoids enumerating all residues of a huge product modulus.

14.3 Beyond quadratic residues

The current sieve uses only the condition that $\Delta(y)$ be a square modulo m . One may ask whether stronger local conditions can be extracted from the full polynomial P , its derivative, or simultaneous congruences in (x, y) . Any such condition must be benchmarked against its random-residue density; otherwise an apparent reduction may simply be a standard local sieve.

14.4 Cross-modulus correlations

Although individual quadratic-residue conditions are largely local, the same y is evaluated against all moduli simultaneously. Measuring correlations between residue classes could reveal whether certain moduli are unusually informative when combined. This should be tested statistically rather than inferred from a small sample.

14.5 Factor-gap information

The relation

$$\Delta(y) = (q - p)^2$$

means that every surviving candidate implicitly predicts a candidate gap $g = \sqrt{\Delta(y)}$. A future route would need a constraint on g that is stronger than local quadratic residuosity but does not amount to simply finding the square. The current 2-adic experiments provide structural information about $v_2(g)$, but not a sufficient selector.

15 Conclusion

The arithmetic-sequence construction developed here leads to a surprisingly rigid algebraic picture. The weighted binomial identity transforms the original quadratic polynomial into a simpler quadratic $H(x)$; diagonalization removes the quadratic term entirely and exposes a parameter $r = s + z$ whose divisibility by N is equivalent to an integer zero on that diagonal. The normalized transformed difference recovers this same parameter exactly.

At the same time, the y -coordinate reveals that the entire polynomial system is a shifted representation of Fermat factorization. The discriminant is exactly $(q - p)^2$, and the alternative (a, b) variables reduce to the same square condition. This prevents overinterpretation of the algebra: several elegant identities describe the factorization problem without making it easier in a complexity-theoretic sense.

The modular analysis is richer. Modulo 2^k , solvability has an exact parity criterion, the solution set carries an involution, its fixed points are shifted square roots of N , and those square-root branches encode the 2-adic valuation of the factor gap. Yet targeted collision experiments rule out the simplest hypothesis that this information is determined by low-bit fingerprints of N and s .

The strongest computational result is the quadratic-residue sieve on the Fermat coordinate. A carefully chosen collection of small moduli can eliminate roughly three orders of magnitude of candidate values while preserving the true factor coordinate. The current evidence, however, indicates that the true coordinate is merely an ordinary survivor of this sieve, not a statistically privileged one.

Thus the research program ends at a useful boundary. The arithmetic-sequence identities, diagonal parameterization, Fermat equivalence, 2-adic involution, and quadratic-residue wheel form a coherent exact framework. What remains is to find a condition that is simultaneously stronger than ordinary local residuosity, cheaper than an exact square test, and genuinely correlated with the hidden factor location. That is the point at which the present algebraic reparameterizations would have to give way to a new factor-discovery mechanism.

A Representative computational verification

The computational experiments used random distinct odd primes and verified identities over the resulting semiprimes. A representative pseudocode skeleton is:

```
def generate_case(bits):
    p = random_odd_prime(bits)
```

```

q = random_odd_prime(bits)
while q == p:
    q = random_odd_prime(bits)
if p > q:
    p, q = q, p
N = p*q
s = isqrt(N)
D = N - s*s
x = s + 1 - p
y = p + q - 2*s - 1
return p, q, N, s, D, x, y

def P(N, s, D, x, y):
    return x*x + (y-1)*x + (D-s) - (s+1)*y

def discriminant(N, s, y):
    return (2*s + 1 + y)**2 - 4*N

def survives_residue_sieve(N, s, y, square_tables):
    delta = discriminant(N, s, y)
    for modulus, table in square_tables.items():
        if table[delta % modulus] == 0:
            return False
    return True

```

For reproducibility, each experiment script printed an explicit beginning and ending marker of the form

```

START EXPERIMENT N
...
FINISHED EXPERIMENT N

```

so that the output streams could be compared unambiguously.

B Notation summary

Symbol	Meaning
N	pq , the odd semiprime
s	$\left\lfloor \sqrt{N} \right\rfloor$
D	$N - s^2$
x	$s + 1 - p$
y	$p + q - 2s - 1$
$P(x, y)$	$x^2 + (y - 1)x + (D - s) - (s + 1)y$
z	$x + y$
r	$s + z$, the diagonal divisor parameter
$H(x)$	$4x^2 - (s + 4)x + 3D - 3s$
A	$s + \lceil y/2 \rceil$, Fermat's square-root coordinate
a	$s - p$
b	$q - s$
v_2	2-adic valuation